

Multiplicity of Eigenvalues

$$\begin{bmatrix} 2 & 0 & 1 \\ -1 & 4 & -1 \\ -1 & 2 & -2 \end{bmatrix}$$

Finding the eigenvalues and eigenvector(s) :

$$\det(A - \lambda I) = 0$$

$$|A - \lambda I| = 0$$

$$\Rightarrow \begin{vmatrix} 2-\lambda & 0 & 1 \\ -1 & 4-\lambda & -1 \\ -1 & 2 & -\lambda \end{vmatrix} = 0$$

$$\Rightarrow (2-\lambda)[(4-\lambda)(-\lambda) - (-1)(2)] \neq 0 + 1[(-1)(2) - (-1)(4-\lambda)]$$

$$(2-\lambda)[(-4\lambda + \lambda^2) + 2] + 1[-2 + 4 - \lambda]$$

$$(2-\lambda)[\lambda^2 - 4\lambda + 2] + 1[2 - \lambda]$$

$$(2-\lambda)(\lambda^2 - 4\lambda + 2) + 1(2 - \lambda)$$

$$(2-\lambda)(\lambda^2 - 4\lambda + 2) + (2 - \lambda)$$

$$(2-\lambda)[(\lambda^2 - 4\lambda + 2) + 1]$$

$$(2-\lambda)[\lambda^2 - 4\lambda + 3]$$

$$(2-\lambda)(\lambda^2 - 4\lambda + 3) = 0$$

$$(2-\lambda)(\lambda^2 - 3\lambda - \lambda + 3) = 0$$

$$(2-\lambda)(\lambda(\lambda-3) - 1(\lambda-3)) = 0$$

$$(2-\lambda)(\lambda-1)(\lambda-3) = 0$$

$$\Rightarrow 2 - \lambda = 0$$

$$\lambda - 1 = 0$$

$$\lambda - 3 = 0$$

$$\underline{\underline{2 = \lambda}}$$

$$\underline{\underline{\lambda = 1}}$$

$$\underline{\underline{\lambda = 3}}$$

For $\lambda = 2$; the corresponding eigenvector :

$$\Rightarrow \begin{bmatrix} 2-2 & 0 & 1 \\ -1 & 4-2 & -1 \\ -1 & 2 & 0-2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 0 & 0 & 1 \\ -1 & 2 & -1 \\ -1 & 2 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow x_3 = 0$$

$$\Rightarrow -x_1 + 2x_2 - x_3 = 0 \Rightarrow -x_1 + 2x_2 - x_3 = 0$$

$$\Rightarrow -x_1 + 2x_2 = 0 \Rightarrow -x_1 + 2x_2 - 2x_3 = 0$$

$$2x_2 = x_1$$

$$\text{Since } x_3 = 0;$$

$$\Rightarrow -x_1 + 2x_2 - 2x_3 = 0 \therefore -x_1 + 2x_2 = 0$$

$$-(2x_2) + 2x_2 = 0 \quad -x_1 + 2x_2 = 0$$

$$-2x_2 + 2x_2 = 0$$

$$\Rightarrow x_1 = 2x_2$$

$$\Rightarrow x_1 = 2x_2$$

$$\therefore \begin{bmatrix} 2x_2 \\ x_2 \\ 0 \end{bmatrix} = x_2 \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} ; \text{ Let } x_2 = 1;$$

$$\therefore \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}$$

① Multiplicity is always associated with the eigenvalue.

$\Rightarrow$ Since the eigenvalues are ; $\lambda_1 = 2$, $\lambda_2 = 1$, $\lambda_3 = 3$;

Thus, the matrix has three distinct eigenvalues. Their algebraic multiplicities are :

$$\Rightarrow \mu(\lambda_1) = 1$$

$$\mu(\lambda_2) = 1$$

$$\mu(\lambda_3) = 1$$

MULTIPLICITY OF Eigenvalues

The **algebraic multiplicity** of an eigenvalue is the number of times it appears as a root of the characteristic polynomial (i.e. the polynomial whose roots are the eigenvalue of a matrix).

e.g. Consider the 2×2 matrix :

$$A = \begin{bmatrix} 4 & 2 \\ 1 & 2 \end{bmatrix}$$

The characteristic polynomial is :

$$\Rightarrow \det(A - \lambda I) = 0$$

$$\Rightarrow \begin{vmatrix} 4-\lambda & 2 \\ 1 & 2-\lambda \end{vmatrix} = 0$$

$$\Rightarrow (4-\lambda)(2-\lambda) - 2 = 0$$

$$8 - 4\lambda - 2\lambda + \lambda^2 - 2 = 0$$

$$\lambda^2 - 6\lambda + 6 = 0$$

$$\therefore \lambda_1 = 3 + \sqrt{3}$$

$$\lambda_2 = 3 - \sqrt{3}$$

□ Thus, A has two distinct eigenvalues. Their algebraic multiplicities are :

$$\Rightarrow \mu(\lambda_1) = 1$$

$$\mu(\lambda_2) = 1,$$

because they are NOT repeated.

e.g. Consider the 2×2 matrix :

$$A = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}$$

The characteristic polynomial is :

$$\Rightarrow \det(A - \lambda I) = 0$$

$$\Rightarrow \begin{vmatrix} 1-\lambda & 0 \\ 2 & 1-\lambda \end{vmatrix} = 0$$

$$\Rightarrow (1-\lambda)(1-\lambda) - 0 = 0$$

$$\Rightarrow (1-\lambda)^2 = 0$$

$$1-\lambda = 0$$

$$\therefore \lambda_1 = 1$$

$$\lambda_2 = 1$$

④ Thus, A has one repeated eigenvalue whose algebraic multiplicity is :

$$\Rightarrow \mu(\lambda_1) = \mu(\lambda_2) = 2$$

Differential Equations

First Order Linear Differential Equations

$$\Rightarrow \frac{dy}{dx} + P(x)y = Q(x)$$

$$\Rightarrow y' + P(x)y = Q(x)$$

$$\Rightarrow \text{Integrating factor} = \int P(x) dx \quad e^{\int P(x) dx}$$

$$\Rightarrow y = \frac{1}{I(x)} \left[\int I(x) Q(x) dx + C \right]$$

where $I(x)$ = Integrating factor

e.g. Solve the differential equation: $\frac{dy}{dx} + 3x^2y = 6x^2$

$$\Rightarrow \frac{dy}{dx} + 3x^2y = 6x^2 \quad ; \quad y' + 3x^2y = 6x^2$$

$$\Rightarrow y' + P(x)y = Q(x)$$

$$\Rightarrow y' + 3x^2y = 6x^2$$

$$\Rightarrow I(x) = e^{\int P(x) dx} = e^{\int 3x^2 dx} = e^{x^3}$$

$$\Rightarrow y = \frac{1}{e^{x^3}} \left[\int e^{x^3} (6x^2) dx + C \right]$$

$$y = \frac{1}{e^{x^3}} \left[\int 6x^2 e^{x^3} dx + C \right]$$

$$y = \frac{1}{e^{x^3}} \left[\frac{6}{3} \int x^2 e^{x^3} dx + C \right]$$

$$y = \frac{1}{e^{x^3}} \left[\frac{6 \cdot 2}{3} (e^{x^3}) + C \right]$$

$$y = \frac{2}{e^{x^3}} (e^{x^3}) + \frac{C}{e^{x^3}} = 2 + \frac{C}{e^{x^3}}$$

$$\therefore y = 2 + Ce^{-x^3}$$

DIFFERENTIAL EQUATIONS

Embedded Derivative Substitutions

e.g. Solve the differential eq. ; $3xy^2 \frac{dy}{dx} = 3x^4 + y^3$

$$\Rightarrow 3xy^2 \frac{dy}{dx} - y^3 = 3x^4 \quad ; \quad \boxed{\text{let } v = y^3}$$

$$\Rightarrow x \left(3y^2 \frac{dy}{dx} \right) - y^3 = 3x^4 \quad \frac{dv}{dx} = 3y^2 \frac{dy}{dx}$$

$$\therefore x \left(\frac{dv}{dx} \right) - v = 3x^4$$

$$\Rightarrow \frac{dv}{dx} - \frac{1}{x}(v) = 3x^3 \quad \left. \vphantom{\frac{dv}{dx}} \right\} \text{bringing to a first order linear differential eq.}$$

$$\Rightarrow v' + \left(-\frac{1}{x} \right) v = 3x^3$$

$$\Rightarrow v' + P(x)v = Q(x)$$

$$\Rightarrow v' + (-1/x)v = 3x^3$$

$$\Rightarrow I(x) = e^{\int P(x)dx} = e^{\int -1/x dx} = e^{-\ln x} = e^{\ln x^{-1}} = x^{-1} = 1/x$$

$$\Rightarrow v = \frac{1}{I(x)} \left[\int I(x) Q(x) dx + c \right] = x \left[\int \left(\frac{1}{x} \right) (3x^3) dx + c \right]$$

$$v = x \left[\int 3x^2 dx + c \right] = x \left[x^3 + c \right] = x^4 + xc$$

$$v = x^4 + xc$$

Since $v = y^3$;

$$\Rightarrow \underline{\underline{y^3 = x(x^3 + c)}}$$

e.g. Solve the differential eq. ; $xe^y \frac{dy}{dx} = 2(e^y + x^3 e^{2x})$

$$\text{Let } v = e^y$$

$$\frac{dv}{dx} = e^y \frac{dy}{dx}$$

$$\Rightarrow x \left(e^y \frac{dy}{dx} \right) = 2(e^y + x^3 e^{2x})$$

$$\Rightarrow x \left(\frac{dv}{dx} \right) = 2(v + x^3 e^{2x})$$

$$\Rightarrow x \frac{dv}{dx} = 2v + 2x^3 e^{2x}$$

$$\Rightarrow x \frac{dv}{dx} - 2v = 2x^3 e^{2x}$$

$$\Rightarrow \frac{dv}{dx} - \frac{2v}{x} = 2x^2 e^{2x}$$

$$\Rightarrow \frac{dv}{dx} - \frac{2}{x}(v) = 2x^2 e^{2x} \quad \left\{ \begin{array}{l} \text{bringing to a first order} \\ \text{Linear differential eq.} \end{array} \right.$$

$$\Rightarrow v' + \left(-\frac{2}{x} \right) v = 2x^2 e^{2x}$$

$$\Rightarrow I(x) = e^{\int P(x) dx} = e^{\int -2/x dx} = e^{-2 \ln x} = e^{\ln x^{-2}} = 1/x^2$$

$$\Rightarrow v = \frac{1}{I(x)} \left[\int I(x) Q(x) dx + c \right] = x^2 \left[\int \left(\frac{1}{x^2} \right) (2x^2 e^{2x}) dx + c \right]$$

$$v = x^2 \left[2 \int e^{2x} dx + c \right] = x^2 \left[2x \left(\frac{1}{2} \right) e^{2x} + c \right] = x^2 e^{2x} + x^2 c$$

$$\Rightarrow v = x^2 (e^{2x} + c)$$

$$\text{Since } v = e^y ;$$

$$\Rightarrow e^y = x^2 (e^{2x} + c)$$

$$\Rightarrow \ln e^y = \ln [x^2 (e^{2x} + c)]$$

$$\therefore y = \ln [x^2 (e^{2x} + c)]$$

Wronskian Determinant

$$\begin{vmatrix} f(x) & g(x) & h(x) \\ f'(x) & g'(x) & h'(x) \\ f''(x) & g''(x) & h''(x) \end{vmatrix} \left. \vphantom{\begin{vmatrix} f(x) & g(x) & h(x) \\ f'(x) & g'(x) & h'(x) \\ f''(x) & g''(x) & h''(x) \end{vmatrix}} \right\} \begin{array}{l} \text{Wronskian} \\ \text{determinant} \end{array} = W$$

If $W \neq 0$, the functions are linearly **independent**.

If $W = 0$, for ODE solutions : linearly **dependent**;
otherwise, test is **inconclusive**.

The Fundamental Definition of Linear Dependence

A set of functions $\{f_1, f_2, \dots, f_n\}$ is linearly dependent on an interval if there exists constants $c_1, c_2, \dots, c_n$ (NOT ALL ZERO) such that :

$$c_1 f_1(x) + c_2 f_2(x) + \dots + c_n f_n(x) = 0$$

for all x in the interval.

① If any function in the set can be expressed as a linear combination of the others, the set is **dependent**.

e.g. Determine whether the given set of functions is linearly independent on the interval $(-\infty, \infty)$.

$$f_1(x) = x, \quad f_2(x) = x^2, \quad f_3(x) = 4x - 3x^2$$

$$\Rightarrow f_3(x) = 4x - 3x^2$$

$$\Rightarrow f_3(x) = 4f_1(x) - 3f_2(x)$$

Since f_3 is a linear combination of f_1 and f_2 , the set is linearly dependent.

② e.g. $f_1(x) = 0, \quad f_2(x) = x, \quad f_3(x) = e^x$

$\Rightarrow$ Any set containing the zero function is automatically linear dependent because 0 can be written as :

$$\Rightarrow f_1(x) = 0$$

$$\Rightarrow f_1(x) = f_1(x)f_2(x) + f_1(x)f_3(x)$$

SECOND-ORDER DIFFERENTIAL equations

When solving a second-order linear homogenous differential equation of the form:

$$ay'' + by' + cy = 0;$$

the characteristic equation is:

$$ar^2 + br + c = 0$$

The nature of the roots r_1 and r_2 , determines the form of the complementary solution y_c . Below are the three possible cases:-

CASE I: Distinct Real Roots ($r_1 \neq r_2$)

Complementary solution:

$$y_c = C_1 e^{r_1 x} + C_2 e^{r_2 x}$$

e.g. For $y'' - 5y' + 6y = 0$, the characteristic equation is

$$r^2 - 5r + 6 = 0; \quad r_1 = 2, 3$$

$$\therefore \text{SOLUTION: } y_c = C_1 e^{2x} + C_2 e^{3x}$$

CASE II: Repeated Real Roots ($r_1 = r_2$)

Complementary solution:

$$y_c = C_1 e^{rx} + C_2 x e^{rx}$$

e.g. For $y'' - 4y' + 4y = 0$, the characteristic equation is

$$r^2 - 4r + 4 = 0; \quad r = 2, 2$$

$$\therefore \text{SOLUTION: } y_c = C_1 e^{2x} + C_2 x e^{2x}$$

CASE III: Complex Conjugate Roots ($r = \alpha \pm ib$)

Complementary solution:

$$y_c = e^{\alpha x} (C_1 \cos(bx) + C_2 \sin(bx))$$

e.g. For $y'' + 4y' + 13y = 0$, the characteristic equation is

$$r^2 + 4r + 13 = 0; \quad r = -2 \pm 3i.$$

$$\therefore \text{SOLUTION: } y_c = e^{-2x} (C_1 \cos(3x) + C_2 \sin(3x))$$

Power Series

$$\sum_{n=0}^{\infty} a_n x^n \Rightarrow \text{centered at } c=0$$

$$\sum_{n=0}^{\infty} a_n (x-c)^n \Rightarrow \text{centered at } c=c$$

In order to determine the **radius** and **interval of convergence**, we use the ratio test:

$$\Rightarrow \lim_{n \rightarrow \infty} \left| \frac{u_{n+1}}{u_n} \right|$$

There are three (3) cases that arise when applying the ratio test:

$$\text{i- } \lim_{n \rightarrow \infty} \left| \frac{u_{n+1}}{u_n} \right| = 0 \quad \left. \vphantom{\lim_{n \rightarrow \infty} \left| \frac{u_{n+1}}{u_n} \right| = 0} \right\} \begin{array}{l} R = \infty \\ I = (-\infty, \infty) \end{array}$$

$$\text{ii- } \lim_{n \rightarrow \infty} \left| \frac{u_{n+1}}{u_n} \right| = \infty \quad \left. \vphantom{\lim_{n \rightarrow \infty} \left| \frac{u_{n+1}}{u_n} \right| = \infty} \right\} \begin{array}{l} R = 0 \\ I = \{c\} \text{ ; where the series} \\ \text{converges at } x=c; \end{array}$$

$$\text{iii- } \lim_{n \rightarrow \infty} \left| \frac{u_{n+1}}{u_n} \right| = \frac{1}{R} \quad |x-c| < 1 = |x-c| < R$$

$$\Rightarrow |x-c| < R \quad \text{the series converges}$$

$$\Rightarrow |x-c| < R$$

$$-R < x-c < R$$

$$-R+c < x < R+c \quad \left. \vphantom{-R+c < x < R+c} \right\} \begin{array}{l} R = R \\ I = (-R+c, R+c) \end{array}$$

$$-R < x-a < R$$

$$-R+a < x < R+a$$

$$|x-a| < \underline{\underline{R}}$$

e.g. Find the interval and radius of convergence for :

$$\sum_{n=1}^{\infty} \frac{(x-3)^n}{2^n n}$$

⇒ Applying the ratio test ;

$$\Rightarrow \lim_{n \rightarrow \infty} \left| \frac{u_{n+1}}{u_n} \right| = \lim_{n \rightarrow \infty} \left| \frac{(x-3)^{n+1}}{2^{n+1} (n+1)} \times \frac{2^n n}{(x-3)^n} \right| < 1$$

$$\Rightarrow \lim_{n \rightarrow \infty} \left| \frac{(x-3)^{n+1-n} 2^{n-n-1} n}{(n+1)} \right| = \lim_{n \rightarrow \infty} \left| \frac{(x-3) 2^{-1} n}{(n+1)} \right|$$

$$\Rightarrow \lim_{n \rightarrow \infty} \left| \frac{(x-3) n}{2 (n+1)} \right| = \lim_{n \rightarrow \infty} \left| \frac{1}{2} (x-3) \right| \cdot \left| \frac{n}{n+1} \right|$$

$$\Rightarrow \frac{1}{2} |x-3| \cdot \lim_{n \rightarrow \infty} \left| \frac{n}{n+1} \right| \rightarrow \text{applying L'Hopital's rule ;}$$

$$\Rightarrow \frac{1}{2} |x-3| \cdot (1) < 1$$

$$\Rightarrow \frac{1}{2} |x-3| < 1 ; |x-3| < 2$$

$$\Rightarrow \text{Radius of convergence} = R = 2$$

$$\Rightarrow \text{Interval of convergence} = |x-3| < 2$$

$$-2 < x-3 < 2$$

$$-2+3 < x < 2+3$$

$$\underline{\underline{1 < x < 5}}$$

SOLVING DIFFERENTIAL EQUATIONS WITH POWER SERIES

$$y'' - 2y = 0$$

$$\Rightarrow y = \sum_{n=0}^{\infty} a_n x^n \quad y' = \sum_{n=1}^{\infty} a_n (n) x^{n-1} \quad y'' = \sum_{n=2}^{\infty} a_n (n)(n-1) x^{n-2}$$

$$\therefore \sum_{n=2}^{\infty} n(n-1) a_n x^{n-2} - x \sum_{n=0}^{\infty} a_n x^n = 0$$

$$\Rightarrow \sum_{n=2}^{\infty} n(n-1) a_n x^{n-2} - \sum_{n=0}^{\infty} a_n x^{n+1} = 0$$

$$\Rightarrow \text{let } k = n-2$$

$$\therefore n = k+2$$

$$\text{let } k = n+1$$

$$n = k-1$$

$$\Rightarrow \sum_{k=0}^{\infty} (k+2)(k+1)a_{k+2}x^k - \sum_{k=1}^{\infty} a_{k-1}x^k = 0$$

$\Rightarrow$ In order to match the index :

$$\Rightarrow \left[(0+2)(0+1)a_2x^0 \right] + \sum_{k=1}^{\infty} (k+2)(k+1)a_{k+2}x^k - \sum_{k=1}^{\infty} a_{k-1}x^k = 0$$

$$\Rightarrow 2a_2 + \sum_{k=1}^{\infty} (k+2)(k+1)a_{k+2}x^k - \sum_{k=1}^{\infty} a_{k-1}x^k = 0$$

$$\Rightarrow 2a_2 + \sum_{k=1}^{\infty} [(k+2)(k+1)a_{k+2} - a_{k-1}]x^k = 0$$

$$\Rightarrow 2a_2 = 0 \quad (k+2)(k+1)a_{k+2} - a_{k-1} = 0$$

$$a_2 = 0$$

$$a_{k+2} = a_{k-1}$$

$$a_2 = 0$$

$$(k+2)(k+1)$$

$$a_3 = a_0/2.3 = a_0/6$$

$$a_4 = a_1/4.3 = a_1/12$$

$$a_5 = a_2/5.4 = 0$$

$$a_6 = a_3/6.5 = (a_0/6)(1/30) = a_0/180$$

$$a_7 = a_4/7.6 = (a_1/12)(1/42) = a_1/504$$

For the two solutions :

$$\Rightarrow \text{Let } a_0 = 1, \quad a_1 = 0$$

$$y_1(x) = 1 + \frac{1}{6}x^3 + \frac{1}{180}x^6 + \frac{1}{12960}x^9 + \dots$$

$$\Rightarrow \text{Let } a_0 = 0, \quad a_1 = 1$$

$$y_2(x) = x + \frac{1}{12}x^4 + \frac{1}{504}x^7 + \dots$$

$$\therefore y_1(x) = a_0 \left[1 + \frac{x^3}{6} + \frac{x^6}{180} + \frac{x^9}{12960} + \dots \right]$$

$$y_2(x) = a_1 \left[x + \frac{x^4}{12} + \frac{x^7}{504} + \dots \right]$$

Q. Find the interval and radius of convergence for the given power series :-

Q1. $\sum_{n=1}^{\infty} \frac{1}{n^2} x^n$

In order to find the interval and radius of convergence, we apply the ratio test :

$$\Rightarrow \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| < 1 ; \lim_{n \rightarrow \infty} \left| \frac{1}{(n+1)^2} x^{n+1} \times \frac{n^2}{x^n} \right| < 1$$

$$\Rightarrow \lim_{n \rightarrow \infty} \left| \frac{x^{n+1-n} n^2}{(n+1)^2} \right| < 1$$

$$\Rightarrow \lim_{n \rightarrow \infty} \left| \frac{x n^2}{(n+1)^2} \right| < 1 ; \lim_{n \rightarrow \infty} \left| \frac{(x) n^2}{(n+1)^2} \right| < 1$$

$$\Rightarrow |x| \cdot \lim_{n \rightarrow \infty} \left| \frac{n^2}{(n+1)^2} \right| < 1 ; |x| \cdot \lim_{n \rightarrow \infty} \left| \left(\frac{n}{n+1} \right)^2 \right| < 1$$

$$\frac{n}{n+1} = \frac{n}{n(1+1/n)} = \frac{1}{1+1/n}$$

$$\therefore \Rightarrow |x| \cdot \lim_{n \rightarrow \infty} \left| \left(\frac{1}{1+1/n} \right)^2 \right| < 1$$

$$\Rightarrow |x| \cdot (1)^2 < 1$$

$$\Rightarrow |x| < 1$$

$$\therefore \text{Radius of convergence} = R = 1$$

$$\text{Interval of convergence} = -1 < x < 1$$

Q2. $\sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{9^n}$

In order to find the interval and radius of convergence, we apply the ratio test :

$$\Rightarrow \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| < 1 ; \lim_{n \rightarrow \infty} \left| \frac{(-1)^{n+1} x^{2(n+1)+1}}{9^{n+1}} \times \frac{9^n}{(-1)^n x^{2n+1}} \right| < 1$$

$$\frac{2(n+1)+1-2n-1}{2n+1+1-2n-1}$$

$$\frac{3-1}{2}$$

$$\Rightarrow \lim_{n \rightarrow \infty} \left| (-1)^{n+1-n} x^2 q^{n-n-1} \right| < 1$$

$$\Rightarrow \lim_{n \rightarrow \infty} \left| (-1) x^2 q^{-1} \right| < 1 ; \lim_{n \rightarrow \infty} \left| \frac{(-1)}{q} x^2 \right| < 1$$

$$\Rightarrow \frac{1}{q} \lim_{n \rightarrow \infty} x^2 < 1 ; \frac{1}{q} (x^2) < 1$$

$$\Rightarrow x^2 < q$$

$$\Rightarrow |x| < 3$$

$$\therefore \text{Radius of convergence} = R = 3$$

$$\text{Interval of convergence} = -3 < x < 3$$

Q. Rewrite the given expression using a single power series whose general term x^k :-

$$\text{(Q1.) } \sum_{n=1}^{\infty} 2n c_n x^{n-1} + \sum_{n=0}^{\infty} 6c_n x^{n+1}$$

$$\text{Let } k = n-1$$

$$k = n+1$$

$$n = k+1$$

$$n = k-1$$

$$\Rightarrow \sum_{k=0}^{\infty} 2(k+1) c_{k+1} x^k + \sum_{k=1}^{\infty} 6c_{k-1} x^k$$

$$\Rightarrow [2(0+1)c_1 x^0] + \sum_{k=1}^{\infty} 2(k+1) c_{k+1} x^k + \sum_{k=1}^{\infty} 6c_{k-1} x^k$$

$$\Rightarrow 2c_1 + \sum_{k=1}^{\infty} [(2k+2)c_{k+1} + 6c_{k-1}] x^k$$

$$\text{(Q2.) } \sum_{n=2}^{\infty} n(n-1) c_n x^{n-2} - 2 \sum_{n=1}^{\infty} n c_n x^n + \sum_{n=0}^{\infty} c_n x^n$$

$$\Rightarrow \sum_{n=2}^{\infty} n(n-1) c_n x^{n-2} - \sum_{n=1}^{\infty} 2n c_n x^n + \sum_{n=0}^{\infty} c_n x^n$$

$$\Rightarrow \text{let } k = n-2$$

$$n = k+2$$

$$k = n$$

$$n = k$$

$$k = n$$

$$n = k$$

$$\Rightarrow \sum_{k=0}^{\infty} (k+2)(k+1)c_{k+2}x^k - \sum_{k=1}^{\infty} 2kc_kx^k + \sum_{k=0}^{\infty} c_kx^k$$

$$\Rightarrow [(0+2)(0+1)c_2x^0] + [c_0x^0] + \sum_{k=1}^{\infty} (k+2)(k+1)c_{k+2}x^k - \sum_{k=1}^{\infty} 2kc_kx^k + \sum_{k=1}^{\infty} c_kx^k$$

$$\Rightarrow 2c_2 + c_0 + \sum_{k=1}^{\infty} [(k+2)(k+1)c_{k+2} - 2kc_k + c_k]x^k$$

$$\Rightarrow 2c_2 + c_0 + \sum_{k=1}^{\infty} [(k+2)(k+1)c_{k+2} - c_k(2k-1)]x^k$$

Q. Find a power series solution $y = \sum_{n=0}^{\infty} c_n x^n$ of the given linear first-order differential eq.

Q1. $y' - 5y = 0$

$$y = \sum_{n=0}^{\infty} a_n x^n \quad y' = \sum_{n=1}^{\infty} a_n n x^{n-1} \quad y'' = \sum_{n=2}^{\infty} a_n n(n-1) x^{n-2}$$

$$\Rightarrow \sum_{n=1}^{\infty} a_n n x^{n-1} - 5 \sum_{n=0}^{\infty} a_n x^n = 0$$

$$\Rightarrow \sum_{n=1}^{\infty} a_n n x^{n-1} - \sum_{n=0}^{\infty} 5a_n x^n = 0$$

$$\text{let } k=n-1 \quad k=n$$

$$n=k+1 \quad n=k$$

$$\Rightarrow \sum_{k=0}^{\infty} a_{k+1}(k+1)x^k - \sum_{k=0}^{\infty} 5a_k x^k = 0$$

$$\Rightarrow \sum_{k=0}^{\infty} [a_{k+1}(k+1) - 5a_k]x^k = 0$$

$$\Rightarrow a_{k+1}(k+1) - 5a_k = 0$$

$$\Rightarrow a_{k+1} = \frac{5a_k}{k+1}$$

$5^n x$

$$k=0 \quad a_1 = 5a_0/1! = 5a_0/1!$$

$$k=1 \quad a_2 = 5a_1/2 = 5/2 (5a_0/1!) = 25a_0/2!$$

$$k=2 \quad a_3 = 5a_2/3 = 5/3 (25a_0/2!) = 125a_0/3!$$

$$k=3 \quad a_4 = 5a_3/4 = 5/4 (125a_0/3!) = 625a_0/4!$$

$$y(x) = a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4$$

$$y(x) = a_0 + \left(\frac{5a_0}{1!}\right)x + \left(\frac{25a_0}{2!}\right)x^2 + \left(\frac{125a_0}{3!}\right)x^3 + \left(\frac{625a_0}{4!}\right)x^4$$

$$y(x) = a_0 \left[1 + \frac{5x}{1!} + \frac{25x^2}{2!} + \frac{125x^3}{3!} + \frac{625x^4}{4!} \right]$$

$$y(x) = a_0 \left[\sum_{k=0}^{\infty} \frac{(5x)^k}{k!} \right]$$

$$\therefore y = a_0 \sum_{k=0}^{\infty} \frac{(5x)^k}{k!}$$
